ERES INSTITUTE FOR NEW AGE CYBERNETICS

Whitepaper

Erasure Completeness Ratio (ECR):

The Falsifiability Metric of MPAM↔MIEVM Duality

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Abstract

The Erasure Completeness Ratio (ECR) is the empirical falsifiability metric of ERES governance, first formalized in the March 2026 SPT specification within the SCALULAR ENGINE document. Canonically, ECR = N_confirmed_destroyed / N_total_propagation_graph: a value in [0, 1] that measures, against a CBGMODD-tracked propagation graph, the fraction of instances of a target object (attestations, permissions, claims, data records, ratified states) that have been verifiably destroyed across every node on which they ever propagated. ECR is bounded above by the Triune Math key REAL = (E · M · R) / (T · S), so ECR → 1 asymptotically and the residual (1 − ECR) must be contained rather than denied. ECR is the closure metric of the MPAM↔MIEVM Duality — the Melting Pot Assimilation Method (affirmative ledger) and the Multi-Instrument Ensemble Validation Method (operational runtime) — with ECR's asymptotic-yet-falsifiable character converting ERES from an abstract narrative system into a governable, empirically testable control law over a one-thousand-year horizon. This whitepaper restates the canonical formula, anchors it within C = R × P ÷ M and the REAL bound, specifies its instrumentation through EAAP and EAAP-CRL under ML-DSA/Falcon post-quantum signatures, fixes the SCALULAR pillar thresholds (HEALTH ≥ 0.99, PROTECTION ≥ 0.97), and demonstrates the seven-vector application across the RATE dimensions $R_1 \ldots R_7$.

1. Background and Provocation

ERES has been challenged — appropriately and repeatedly — on whether its symbolic apparatus is empirically validatable, or whether the equations, backronyms, and lock-boxes constitute an abstract narrative system indistinguishable from rhetoric. The most recent and most precise form of this challenge, raised in May 2026 by an Execution Governance Strategist working on runtime oversight for autonomous AI systems, is reproduced in paraphrase:

> *The issue is less the symbolic equation itself and more whether the underlying governance logic can be empirically validated. Without definitional clarity around the variables and control structure, these models become difficult to distinguish from abstract narrative systems.*

This challenge is legitimate. It is also, within ERES, already answered — but the answer requires naming the specific instrument that closes the loop, rather than restating the equation. That instrument is ECR. The remainder of this document specifies it.

2. The Canonical Root

ECR cannot be defined in isolation. It is the falsifiability metric of a specific control law, and that control law must be stated first, in its canonical form, with no variable migration.

2.1 Triune Math Key #1

$$\mathbf{C = R \times P \div M}$$

Cybernetics = Resource × Purpose ÷ Method

The variables are fixed. Drift in any of them — substituting Consequence for Cybernetics, Permissions for Purpose, Mitigation for Method — produces a different equation wearing the same letters, and is itself a failure mode (repeatable drift) that ECR is partly designed to detect across instruments.

2.2 The Three Governing Principles

Every ECR computation is subordinated to:

1. Don’t hurt yourself.
2. Don’t hurt others.
3. Build for generations to come.

An ECR of 1.0 obtained at the cost of violating any of these three is, by ERES doctrine, a null result.

2.3 Administration

C = R × P ÷ M is administered by the User-GROUP Community of Interest (COI) operating through GAIA. GAIA carries a deliberate dual reference: as institutional backronym it names the Global Actuary Investor Authority — the federated governance body responsible for implementing ERES standards and overseeing planetary resource allocation as the actuary of the planet; as substrate referent it names the biosphere itself — the planet as audit ground on which the institutional GAIA must report. HowWay is its operational handle. ECR is the number that tells the COI whether HowWay is actually doing what HowWay claims to do, with the actuarial GAIA reporting to and constrained by the biospheric GAIA.

3. The MPAM↔MIEVM Duality

Every ERES governance act has two faces:

MPAM (affirmative side)

Melting Pot Assimilation Method — the formal modular protocol by which mathematical frameworks, attestations, permissions, claims, and ratified states are admitted into the ERES corpus only when needed, as needed, and where needed. MPAM establishes the layered architecture, trigger-based adoption rules, containment tiers, and provenance-locking requirements that constitute the corpus of things ERES has positively assimilated. In the governance frame addressed by this whitepaper, MPAM is the affirmative ledger of what has been said yes to.

MIEVM (validation side)

Multi-Instrument Ensemble Validation Method — the cross-instrument check, currently realized through a short-list ensemble (Claude, Grok, DeepSeek, ChatGPT) culled by bandwidth constraint from a larger original set. MIEVM is the operational runtime of MPAM: it tests whether material admitted under MPAM holds up under independent re-derivation by multiple instruments. Where MPAM specifies what may be assimilated, MIEVM determines whether the assimilation actually converged.

The duality claim

MPAM is only as trustworthy as MIEVM can re-derive it, and MIEVM is only as trustworthy as the erasure side of MPAM is complete. A framework, attestation, or claim assimilated under MPAM that cannot be cleanly de-assimilated when its triggers fail is a framework that was never really under control. The duality is therefore not a redundancy — it is a closure requirement, with de-assimilation (Tier 5 Archive) as the structural mirror of assimilation. ECR is the metric of that closure.

4. Erasure Completeness Ratio — Formal Definition

4.1 The canonical ratio

The Erasure Completeness Ratio was first formalized in the March 2026 SPT specification within the SCALULAR ENGINE document, in the context of CBGMODD's reframed role as erasure-completeness auditor rather than binary-certificate issuer. The canonical formula is:

$$\mathbf{ECR = N\_confirmed\_destroyed \div N\_total\_propagation\_graph}$$

Where N_confirmed_destroyed is the number of confirmed-eliminated instances of the target object (including copies, caches, backups, derivative products, signed attestations, granted permissions, and ratified claim states) and N_total_propagation_graph is the total number of instances identified by CBGMODD's provenance tracking across the object's full lifecycle. ECR is a value in [0, 1].

4.2 The REAL bound

ECR is structurally bounded by the third Triune Math key, REAL = (E · M · R) / (T · S):

$$\mathbf{ECR\_max = f(E, M, R, T, S)}$$

Erasure effectiveness is bounded by the Energy and Matter resources that can be applied, modulated by the Resonance of the system's coordination capacity, across the Time and Space over which the object has propagated. This is the same thermodynamic frontier that bounds complete recycling of matter — there is always residual signal (magnetic remnance, unenumerated cache fragments, derivative hashes, neural traces in human readers, propagated copies in surfaces not yet discovered by CBGMODD). The honest engineering position is therefore not "ECR can reach 1" but rather "ECR approaches 1 asymptotically and the residual is what we contain." When the marginal cost of pursuing the next data fragment exceeds the REAL value of the system, the practical limit of erasure has been reached for that window.

4.3 Interpretation

- **ECR → 1.0** — asymptotic ceiling; full closure within the resource envelope; residual is bounded by REAL.
- **ECR < 1.0** — residual entries persist beyond the REAL frontier: stale signatures still verifying, revoked permissions still being honored, retracted claims still propagating downstream. The complement (1 − ECR) is the leakage fraction; it must be contained, not denied.
- **ECR = VEILED** — reserved state for windows in which the propagation graph itself is unresolvable (the denominator cannot be enumerated, typically because CBGMODD has not yet mapped a surface that may hold residual copies). VEILED replaces the earlier ⊥ (bottom) notation; it is not zero, not undefined, and not failure — it is the explicit acknowledgement that the question is not currently answerable, which itself is information.

4.4 The seven RATE dimensions

ECR is not a single scalar in production deployments. Following the canonical relation (CBGMODD × FAVORS) + BERA = RATE, which yields seven RATE dimensions R_1 through R_7, a production ECR is reported as the vector $\langle ECR_1, ECR_2, \ldots, ECR_7 \rangle$ (one ratio per RATE dimension), and the convergent mean across the seven is what is compared to the MIEVM GO threshold (≥ 8.5). A scalar reported without the seven-vector behind it is presumptively VEILED. This is the whitepaper's synthesis: the canonical ECR formula applied dimension-wise across RATE.

4.5 Why this is empirically testable

Each term in the canonical formula is operationally defined and externally measurable:

- N_total_propagation_graph is enumerated by CBGMODD's provenance tracking against the COI charter and the EAAP-CRL revocation list — all public, all signed, all auditable.
- N_confirmed_destroyed is determined by polling every node in the propagation graph (registries, mirrors, cached attestations, replica nodes, derivative products) and confirming that the target instance is no longer present or that signatures over it no longer verify.
- The numerator is a subset of the denominator by construction; both are finite within the operational window.

A narrative claim that "the system retires obsolete permissions" cannot be falsified. ECR can be falsified, in principle, by any third party with access to the EAAP-CRL, the CBGMODD propagation graph, and the downstream surfaces; if their independent computation diverges from the ERES-reported ECR, the discrepancy is itself a measurable error term. This is the property that converts the system from narrative to control law.

4.6 SCALULAR pillar thresholds

ECR floors are not uniform across the four SCALULAR pillars; each pillar carries the threshold appropriate to the harm domain it governs:

- **HEALTH (SSHP):** ECR ≥ 0.99 — patient data, medical records, and bio-identifying material require the highest erasure floor.
- **PROTECTION (SSPS):** ECR ≥ 0.97, adjusted downward for inferential-reconstruction risk — whistleblower identity, witness protection, and fear-of-retribution elimination cases.
- **LAW (SSLA) and SKILLS_TRADE (SSST):** thresholds set by the adopting jurisdiction within the COI charter, never lower than 0.90 for any data class touching personally-identifying material.

These thresholds anchor the SCALULAR ENGINE's fear-of-retribution elimination requirement: protecting a disclosure is only credible if the identity of the discloser can be measurably erased to the pillar-appropriate floor.

4.7 Denominator discovery and non-cooperative nodes

The canonical formula presupposes that N_total_propagation_graph is enumerable. In practice, CBGMODD's provenance tracking may be incomplete — some surfaces are partition-resistant (darknet caches, air-gapped replicas, jurisdictionally hostile mirrors), and some nodes may decline to disclose their instance set. v1.2 introduces an explicit discovery protocol and a non-cooperative-node clause closing this gap.

Discovery protocol

CBGMODD performs a two-stage enumeration: (a) declared nodes from the COI charter and EAAP-CRL routing record; (b) sampled discovery against the wider network using a stratified sampling strategy over jurisdictional, technological, and temporal strata. The estimator N̂_total is reported with a 95% confidence interval; N_total_propagation_graph in the published ECR uses the upper bound of that interval. This makes the denominator conservatively large, biasing the reported ECR downward rather than upward — the honest direction of error.

Non-cooperative-node clause

A node that refuses to disclose its instance set, or that cannot be polled within the operational window, is treated by default as a residual contribution to (1 − ECR) until validated by a COI-ordered forensic audit. Persistent refusal across two consecutive windows promotes the affected object's ECR to VEILED with a non-cooperative-node flag, and triggers escalation under §4.8. This means an adversary cannot improve the apparent ECR by hiding propagation; concealment is mathematically equivalent to admitted leakage.

4.8 VEILED governance

VEILED is not a stable resting state. v1.2 specifies entry, residency, and exit criteria so that VEILED cannot mask uncertainty indefinitely.

- **Entry:** an object enters VEILED when (a) CBGMODD cannot enumerate at least one surface in its propagation graph, (b) a node returns non-cooperative for two consecutive windows, or (c) MIEVM pairwise disagreement exceeds the §6.4 bound.
- **Residency:** default time-box of 30 days, extendable once by COI vote to 90 days with published reasoning. Residency time is itself a published audit field.
- **Exit (resolution):** the propagation graph is closed (numeric ECR restored), or remediation completes against the non-cooperative node, or MIEVM re-converges.
- **Exit (failure):** if residency expires without resolution, the object is automatically downgraded — its ECR floor is treated as the pessimistic lower bound (assume all unresolved instances are residual), the affected SCALULAR pillar service is held in CONTAINED state, and a §6.5 drift alarm is raised against the COI.

VEILED therefore has consequences. It is information, not refuge.

5. Instrumentation

ECR is not a calculation performed in isolation; it sits on a stack of named ERES instruments, each of which is itself externally inspectable. The table below maps each sub-component of the ECR computation to the instrument that produces it and to the mode by which a third party can falsify the reported value.

ECR sub-component	ERES instrument	Falsifiability mode
N_total_propagation_graph enumeration	CBGMODD provenance tracking + EAAP-CRL revocation list (post-quantum signed)	Third-party verifies CRL signature, recomputes the propagation graph independently
N_confirmed_destroyed verification	Downstream node polling under ML-DSA / Falcon signatures	Independent re-verification of instance absence or signature non-validity
Subject anchoring	FAVORS six-factor biometric (Fingerprint, Aura, Voice, ODOR, Retina, Signature)	Per-subject erasure traceable to a unique embodied claimant
Erasure-completeness audit (not binary certificate)	CBGMODD acting as completeness auditor; outputs ECR with timestamp	Audit log is signed and re-derivable by any COI member
Cross-instrument check	MIEVM ensemble (Claude, Grok, DeepSeek, ChatGPT) at convergent mean ≥ 8.5	Divergent ECR across instruments → repeatable-drift flag
Tuning signal	BERA Resonance Metrics: ARI (Aura Resonance Index), ERI (Emission Resonance Index), RHC (Resonant Harmony Cycle), RCI (Resonant Continuity Index)	ECR feeds Meritcoin under Proof-of-Resonance (“tuning, not mining”)
Thermodynamic ceiling	REAL = (E · M · R) / (T · S) — Triune Math Key #3	ECR_max = f(E, M, R, T, S); the residual (1 − ECR) is contained, not denied
Unresolvable state	VEILED (supersedes ⊥)	Explicit non-answer; itself audit information

Each row is independently auditable. The full ECR cannot be fabricated without simultaneously fabricating every instrument in the right-hand column — which is precisely the property that converts the system from narrative to control law.

6. ECR as the Falsifiability Hook

The provocation in Section 1 reduces, formally, to: "provide an operationally defined quantity Q such that an external party can compute Q′ from the same inputs and compare Q′ to ERES-reported Q." ECR is that quantity.

6.1 The independent re-derivation test

For any operational window, an external auditor with access to:

- the CBGMODD propagation graph for the target object (the denominator N_total_propagation_graph),
- the published EAAP-CRL for the window,
- the set of downstream nodes declared in the COI charter,
- the post-quantum verification keys (ML-DSA / Falcon public keys),

can compute their own ECR′ = N′_confirmed_destroyed / N′_total_propagation_graph without any privileged cooperation from ERES. The discrepancy $\Delta = |ECR - ECR'|$ is itself a measurable error term, and a non-zero Δ is a falsification event — either an ERES reporting error, an auditor methodology error, or a substrate compromise. All three are actionable, and the partitioning of Δ across the seven RATE dimensions usually identifies which.

6.2 The repeatable-drift detector

A particularly important class of failure that ECR detects: repeatable drift. If across multiple windows $W_1, W_2, \ldots, W_{\square}$ the seven-vector $\langle ECR_1 \ldots ECR_7 \rangle$ trends monotonically downward in any dimension without an explanatory event, the system is leaking erasure in a structured way — quietly accumulating residual permissions or stale attestations. This is the AI-determinism failure mode (deterministic but drifting) reframed in governance terms, and it is distinct from ordinary random error. ECR makes it visible.

6.3 The narrative-system exclusion

A pure narrative system cannot produce an ECR because it has no enumerable propagation graph. The denominator does not exist. Any system that publishes an ECR (with the canonical formula, the seven-vector, and the CBGMODD propagation provenance) has, by that publication alone, exited the narrative-system category. ECR is therefore the bright line.

6.4 MIEVM pairwise disagreement bound

Convergent-mean reporting can mask significant pairwise disagreement among instruments. v1.2 specifies a pairwise bound to close this gap:

$$\max_{i,\square} |ECR_i - ECR_{\square}| \le 0.07$$

If the maximum pairwise divergence across the MIEVM ensemble exceeds 0.07 (seven percentage points), the reported ECR is automatically promoted to VEILED, the full divergence matrix is published, and the responsible instruments are flagged for re-derivation. This rule prevents two instruments from agreeing closely while two others disagree sharply — a configuration that yields an acceptable mean but conceals genuine epistemic disagreement. The 0.07 figure was selected to align with the SCALULAR PROTECTION pillar floor distance ($1.0 - 0.97 = 0.03$) plus the HEALTH-pillar tolerance (0.01) plus a methodological safety margin (0.03); future MIEVM rounds may tighten or relax this bound with explicit COI ratification.

6.5 Drift alarm control limits

Beyond single-window verification, v1.2 specifies pre-registered control limits for the ECR time series, so that drift is detected algorithmically rather than discretionally:

- **Level-1 advisory:** any single-window ECR_i drops below the SCALULAR pillar floor; service for that object class is held in WATCH state pending the next window.
- **Level-2 warning:** monotone downward trend in any ECR_i across three consecutive windows without an explanatory event in the EAAP-CRL log; mandatory COI review.
- **Level-3 alarm:** the convergent mean across all seven RATE dimensions drops below the MIEVM GO threshold (≥ 8.5 normalized), or any $\Delta = |ECR - ECR'|$ exceeds 0.05 against an external auditor; mandatory CONTAINED-state hold on the affected SCALULAR pillar service until remediated.
- **Level-4 escalation:** two consecutive Level-3 alarms within the same operational period escalate to the 1000-year Equity Covenant trustees with public publication of the divergence record.

Control limits are pre-registered, not selected post hoc. This is the property that converts ECR drift from a narrative observation into an actionable signal.

7. ECR Under Post-Quantum Conditions

ECR is meaningful only as long as the signatures it depends on remain meaningful. EAAP migrated to ML-DSA and Falcon in early 2026 specifically to ensure that erasure semantics remain falsifiable on a one-thousand-year horizon (per the Equity Covenant, ERES-COVENANT-2026-001).

Under classical signature schemes, a sufficiently advanced adversary could forge revocation entries or, conversely, forge proofs that revoked objects remain valid. Either attack collapses ECR — the metric returns plausible-looking numbers while the underlying erasure surface is being silently rewritten. ML-DSA (for general attestation) and Falcon (for size-constrained CRL entries) push the falsifiability horizon out beyond foreseeable cryptographic advances, including those leveraging quantum computation.

This is not theoretical hardening. It is the load-bearing requirement for ECR to remain a 1000-year metric rather than a 25-year metric.

8. ECR and Proof-of-Resonance

Meritcoin operates under Proof-of-Resonance, not Proof-of-Work. The distinction matters for ECR specifically:

Proof-of-Work has no native erasure semantics

Once a block is buried under sufficient subsequent work, it is buried. There is no operational notion of "retiring" that block — the chain treats it as permanent. An ECR-equivalent computed against a PoW substrate would always be near zero in the denominator-defined sense (almost nothing is ever required to be erased) and therefore vacuous.

Proof-of-Resonance is continuously revisable

Resonance state is a tuning, not a settlement. The four BERA Resonance Metrics — ARI (Aura Resonance Index, the personal/biometric coherence field), ERI (Emission Resonance Index, the environmental/ecological output measurement), RHC (Resonant Harmony Cycle, the cyclical feedback and temporal rhythm), and RCI (Resonant Continuity Index, the cross-scale persistence metric, formalized in collaboration with Jimmy D. Butzbach's Continuity Theory as RCI = P_Ω_norm × ARI_sys × VibConst) — are re-evaluated at every Meritcoin update window. A prior tuning that no longer holds is not buried; it is retired, and the retirement is exactly what ECR measures. "It's not mining — it's tuning" is therefore not a slogan but a structural claim about the erasure surface: every Meritcoin update presupposes that the erasure of superseded resonance states has been audited by ECR.

ECR is the metric that audits whether the tuning system is honoring its own retirement obligations. Without ECR, Proof-of-Resonance would have the same auditability gap as Proof-of-Work; with ECR, it has the closure property that PoW lacks by construction.

9. The One-Thousand-Year Horizon

The ERES Equity Covenant (ERES-COVENANT-2026-001) is a one-thousand-year governance instrument. ECR is the metric that makes that horizon empirically meaningful rather than aspirational. Three properties together produce this:

4. Each ECR computation is bounded in a specific window W; the horizon is composed of contiguous windows, each independently verifiable.

5. The instrumentation (EAAP-CRL under ML-DSA/Falcon, FAVORS biometric anchoring, MIEVM cross-instrument check) is designed to remain falsifiable across substrate generations — cryptographic, computational, and institutional.
6. VEILED windows are explicitly preserved as VEILED, not retroactively coerced into numerical values. A 1000-year record will contain VEILED entries; their preservation is itself part of the audit substrate.

The horizon, in other words, is not maintained by claiming completeness; it is maintained by maintaining the apparatus that makes incompleteness visible.

10. MIEVM Validation Protocol for This Whitepaper

In accordance with the ERES Force-Multiplier Method v1.5, this document is subject to MIEVM cross-instrument validation prior to canonical deposit. The GO threshold is a convergent mean ≥ 8.5 across the four instruments (Claude, Grok, DeepSeek, ChatGPT), with attention to:

- Canonical preservation: variables in $C = R \times P \div M$ must not migrate.
- Subject-First Nomenclature: ECR is named for what it measures (erasure completeness), not for what it serves.
- Falsifiability: every numerical claim must be reproducible by an external party from declared inputs.
- Drift detection: the seven-vector $\langle ECR_1 \ldots ECR_7 \rangle$ across windows must be preserved, not collapsed to scalar.

Convergence below threshold triggers revision. VEILED is permitted as an instrument response and does not by itself block deposit, provided the VEILED reasoning is itself reported.

11. Worked Example — Whistleblower Disclosure Under SCALULAR PROTECTION

This section makes the canonical formula concrete with a single end-to-end example. All numbers are illustrative; the structure is canonical.

11.1 Scenario

Operational window: W = 2026-Q2, length 90 days. Target object: personally-identifying information (PII) associated with whistleblower disclosure ID-WB-2026-042, governed under the SCALULAR PROTECTION pillar (SSPS, floor ECR ≥ 0.97 with inferential-reconstruction adjustment). At the start of W, the COI has issued an EAAP-CRL entry revoking all attestations referencing the discloser’s identity; CBGMODD begins propagation-graph enumeration.

11.2 Propagation-graph enumeration

CBGMODD enumerates 47 declared nodes carrying the target object:

- 12 EAAP attestation surfaces (primary credentials)
- 8 cached attestation mirrors (distributed across jurisdictions)
- 14 downstream replica nodes (jurisdictional partner registries)
- 7 derivative-product surfaces (audit logs containing PII references)
- 6 backup vaults (operated by the COI itself)

Stratified discovery sampling (per §4.7) yields an additional 2 nodes not in the declared list — unenumerated cached mirrors at non-COI infrastructure providers. CBGMODD updates the denominator: N_total_propagation_graph = 49 (upper bound of the 95% confidence interval).

11.3 Erasure cascade and ECR computation

Over W, revocation completes across 45 nodes. Two backup vaults are on a scheduled erasure cycle that extends into W+1, and the two newly-discovered unenumerated mirrors return non-cooperative responses to polling. Under the §4.7 clause, the two non-cooperative mirrors are treated as residual contributions to (1 − ECR).

$$\mathbf{ECR(W) = 45 / 49 = 0.918}$$

This is below the PROTECTION floor of 0.97. The PROTECTION pillar service for ID-WB-2026-042 is held in CONTAINED state. The two non-cooperative mirrors are flagged under §4.7; remediation is initiated through jurisdictional channels.

11.4 The independent re-derivation

An external auditor with access to the CBGMODD propagation graph, the EAAP-CRL, and the ML-DSA / Falcon verification keys recomputes against the same surfaces and discovers one additional jurisdictional replica node that CBGMODD had not enumerated. Their computation:

$$\mathbf{ECR'(W) = 45 / 50 = 0.900}$$

$$\mathbf{\Delta = |0.918 - 0.900| = 0.018}$$

Δ is below the §6.5 Level-3 alarm threshold of 0.05, but it is a measurable error term. The auditor publishes their finding; CBGMODD ingests the additional node into its propagation graph; the canonical ECR is updated to 45/50 = 0.900. The PROTECTION pillar remains in CONTAINED state.

11.5 The VEILED case

One of the two non-cooperative mirrors is located in a partition-resistant darknet replica. The set of instances at that mirror is not enumerable by external means. The sub-graph terminating at that node is reported as VEILED under §4.8 with reason “non-enumerable mirror at node

ND-4127." Default residency: 30 days. COI extends to 90 days by published vote, citing diplomatic remediation in progress. If unresolved at day 90, the object's ECR is automatically downgraded to the pessimistic lower bound (assume all unresolved instances are residual), and a §6.5 Level-3 drift alarm is raised against the COI.

11.6 The MIEVM cross-check

Four instruments independently compute ECR for ID-WB-2026-042 over W:

- Claude: 0.918
- ChatGPT: 0.910
- DeepSeek: 0.925
- USE.ai: 0.915

Convergent mean = 0.917. Max pairwise divergence = |0.925 − 0.910| = 0.015, which is below the §6.4 bound of 0.07. MIEVM converges; the reported ECR is canonical at 0.918 (Claude's value adopted as the publishing instrument), with the four-instrument record published as part of the audit trail. If max pairwise had exceeded 0.07, the ECR would have been promoted to VEILED with the full divergence matrix published.

11.7 What the example demonstrates

Every claim in this worked example is independently re-derivable from the published CBGMODD propagation graph, the EAAP-CRL for W, the ML-DSA / Falcon verification keys, and the COI charter. There is no step that depends on privileged ERES cooperation. The example is therefore itself an instance of the property the whitepaper claims for ECR: a third party can verify or falsify any number in this section without asking ERES for permission.

12. Reference Implementation Roadmap

Independent re-derivation requires public tooling. v1.2 commits to the following sequence of open releases through 2027, each milestone gated by MIEVM convergence and a published audit record.

12.1 Phase 1 — CBGMODD enumerator open release (2026-Q3)

Open-source release of the CBGMODD provenance-tracking enumerator with a synthetic test corpus, EAAP-CRL integration under ML-DSA, and a worked specification of the discovery-sampling estimator. Reproduction kit will allow third parties to enumerate propagation graphs for arbitrary attested objects against a public test substrate.

12.2 Phase 2 — Public ECR testnet (2026-Q4)

Public testnet hosting synthetic propagation graphs of escalating complexity (10^2, 10^4, and 10^6 nodes). Third parties run their own ECR′ computation; the testnet publishes the Δ leaderboard. Convergence below Δ = 0.01 across at least three independent implementations gates Phase 3.

12.3 Phase 3 — Absence-proof primitives (2027-Q1)

Integration of authenticated data-structure primitives (Merkle-tree commitments per RFC 6962, verifiable transparency logs, succinct non-membership proofs) so that N_confirmed_destroyed claims are themselves cryptographically attestable rather than asserted. This is the load-bearing requirement for distinguishing “confirmed destroyed” from “asserted destroyed.”

12.4 Phase 4 — First production audit (2027-Q2)

First production audit against the ERES-MPAM-2026-001 corpus using the open tooling. ECR computed by three independent implementations; results published with full divergence matrix and Δ analysis. This is the empirical milestone that promotes the framework from specified to demonstrated.

12.5 Phase 5 — SCALULAR HEALTH pillar pilot (2027-Q3)

First operational SCALULAR pillar deployment, targeting HEALTH (SSHP) given its 0.99 floor and the relative maturity of medical-record erasure norms (HIPAA-aligned). Pilot includes published cost-risk model (per Appendix D) and continuous public ECR reporting.

12.6 Open dependencies

The roadmap depends on three things ERES cannot fund alone: an independent auditor willing to run the Phase 2 testnet, jurisdictional partners willing to host Phase 4 propagation surfaces, and a medical-systems integrator willing to host the Phase 5 pilot. v1.2 publishes the roadmap to make these dependencies visible; partner inquiries should reference ERES-ECR-2026-001 v1.2.

13. Conclusion

The Erasure Completeness Ratio is the specific empirical instrument that converts ERES governance from a symbolic apparatus into a falsifiable control law. It does this by making one question operationally answerable across any deployment window and any governance object class:

“When this system says something is gone, is it actually gone, everywhere?”

ECR returns a number in [0, 1] computed against a CBGMODD-tracked propagation graph and bounded above by REAL, or it returns VEILED with documented entry/exit criteria and a 30-day default residency. Either return is information. Either return is auditable. Either return is independently re-derivable from the published surface.

v1.2 adds the operational closures the MIEVM ensemble identified as gaps in v1.1: a discovery-sampling estimator with conservative upper-bound denominators, a non-cooperative-node clause that converts concealment into admitted leakage, time-bounded VEILED governance with automatic downgrade on residency expiry, a pairwise MIEVM disagreement bound, pre-registered drift control limits across four escalation levels, a fully worked example with concrete numbers and an explicit Δ divergence case, and a five-phase reference implementation roadmap through 2027.

C = R × P ÷ M holds. Administered by User-GROUP COI through GAIA — institutional Global Actuary Investor Authority reporting to biospheric audit ground — with HowWay as the operational handle. ECR is the metric by which HowWay demonstrates, window by window, that it is doing what it says it does. The instrument is specified, the falsifiability surface is published, and the implementation path is dated.

Appendix A — Canonical Cross-References

- ERES Triune Math Keys: (1) C = R × P ÷ M (Cybernetics = Resource × Purpose ÷ Method); (2) M × E + C = R; (3) REAL = (E · M · R) / (T · S).
- MPAM — Melting Pot Assimilation Method (formal modular protocol for admitting frameworks/attestations into ERES). Companion document: ERES-MPAM-2026-001.
- MIEVM — Multi-Instrument Ensemble Validation Method (operational runtime of MPAM; current short-list ensemble: Claude, Grok, DeepSeek, ChatGPT; GO threshold convergent mean ≥ 8.5).
- GAIA — dual reference: institutional backronym Global Actuary Investor Authority (federated governance body); substrate referent the biosphere itself (audit ground).
- ERES Equity Covenant — ERES-COVENANT-2026-001 (1000-year governance instrument).
- ERES Attestation and Authorization Protocol — EAAP v1.5, with EAAP-CRL under ML-DSA / Falcon.
- ERES Reliability Whitepaper — ERES-RELIABILITY-2026-001 (five AI failure vectors mapped to ERES instruments).
- ERES Force-Multiplier Method v1.5 — Subject-First Nomenclature; MIEVM GO threshold convergent mean ≥ 8.5.
- ERES Infomediary (Volume Zero of the ERES Trilogy Architecture), consolidated v2.0.
- SCALULAR Engine Document — four pillars: HEALTH (SSHP, ECR floor ≥ 0.99), LAW (SSLA), PROTECTION (SSPS, ECR floor ≥ 0.97 with inferential-reconstruction

adjustment), SKILLS_TRADE (SSST); Tier 1 = SSSC (Solid-State Smart-City Citizen). Original locus of the canonical ECR formula in the SPT specification (March 2026).

- UBIMIA Manual v1.1 (25 April 2026) — partial instruction set for the PlayNAC engine.
- FAVORS six-factor biometric: Fingerprint · Aura · Voice · ODOR · Retina · Signature.
- BERA Resonance Metrics: ARI (Aura Resonance Index) · ERI (Emission Resonance Index) · RHC (Resonant Harmony Cycle) · RCI (Resonant Continuity Index, with Butzbach formula RCI = P_Ω_norm × ARI_sys × VibConst).
- CBGMODD: Citizen · Business · Government · Military · Ombudsman · Dignitary · Diplomat (seventh seat = Diplomat, not repeated Dignitary). CBGMODD acts as erasure-completeness auditor (not binary-certificate issuer) in the ECR context.
- (CBGMODD × FAVORS) + BERA = RATE, yielding seven RATE dimensions R_1 through R_7.
- VEILED — supersedes ⊥ (bottom) as the semantically unresolvable RATE state.

Appendix B — Notation

- **ECR = N_confirmed_destroyed / N_total_propagation_graph** — canonical Erasure Completeness Ratio (SPT specification, March 2026).
- **ECR_max = f(E, M, R, T, S)** — REAL bound on ECR; the thermodynamic ceiling derived from Triune Math Key #3.
- ⟨**ECR_1 … ECR_7**⟩ — ECR seven-vector across the RATE dimensions $R_1 \ldots R_7$ (whitepaper synthesis applying the canonical formula dimension-wise).
- **1 − ECR** — leakage fraction; the residual that must be contained rather than denied.
- **VEILED** — explicit unresolvable state; replaces ⊥.
- **Δ = |ECR − ECR′|** — audit divergence between ERES-reported and externally-recomputed ECR.

Appendix C — Version History

v1.0 → v1.1 (canonical-correction pass)

- Corrected MPAM expansion: “Multi-Party Attestation Matrix” → Melting Pot Assimilation Method (canonical, per ERES-MPAM-2026-001).
- Corrected MIEVM expansion: “Multi-Instrument Ensemble Validation Mean” → Multi-Instrument Ensemble Validation Method.
- Corrected ARI expansion: “Aggregate Resonance Index” → Aura Resonance Index.

- Corrected ERI expansion: "Ethical Resonance Index" → Emission Resonance Index.
- Corrected RCI expansion: "Relational Coherence Index" → Resonant Continuity Index, with Butzbach formula added.
- Replaced improvised ECR formula with the canonical SPT formulation: ECR = N_confirmed_destroyed / N_total_propagation_graph.
- Added the REAL bound ECR_max = f(E, M, R, T, S) and the asymptotic-ceiling framing (ECR → 1, never reaches 1).
- Added SCALULAR pillar ECR thresholds (HEALTH ≥ 0.99, PROTECTION ≥ 0.97).
- Reframed CBGMODD as erasure-completeness auditor (not binary-certificate issuer).
- Added dual GAIA framing (Global Actuary Investor Authority + biospheric substrate).

v1.1 → v1.2 (MIEVM Round 1: Claude / ChatGPT / DeepSeek / USE.ai)

- §4.7 (new): Denominator-discovery protocol with stratified sampling; conservative-upper-bound denominator.
- §4.7 (new): Non-cooperative-node clause — concealment is mathematically equivalent to admitted leakage.
- §4.8 (new): VEILED governance with explicit entry, residency (30-day default, extendable to 90 by COI vote), and exit criteria; automatic downgrade on residency expiry.
- §6.4 (new): MIEVM pairwise disagreement bound: $\max_{i,j} |ECR_i - ECR_j| \le 0.07$ else VEILED.
- §6.5 (new): Drift alarm control limits, pre-registered across four escalation levels (advisory, warning, alarm, escalation).
- §11 (new): Full worked example — whistleblower disclosure under SCALULAR PROTECTION, with concrete ECR computation, independent re-derivation, Δ case, VEILED case, and MIEVM cross-check.
- §12 (new): Five-phase reference implementation roadmap through 2027-Q3.
- Appendix D (new): Threshold derivation sketch — cost-risk model for HEALTH (≥ 0.99) and PROTECTION (≥ 0.97) floors.
- Softened the most overtly doctrinal phrasings (per DeepSeek review) without altering canonical formulations.

Appendix D — Threshold Derivation Sketch

The SCALULAR pillar floors (HEALTH ≥ 0.99, PROTECTION ≥ 0.97) are not arbitrary. v1.2 provides a brief cost-risk derivation; a full quantitative model will accompany the Phase 5 pilot per §12.5.

D.1 HEALTH (SSHP, floor ≥ 0.99)

HEALTH governs medical PII, bio-identifying material, and patient-record erasure. The expected harm per residual record (E[H]) integrates regulatory penalties (HIPAA-aligned baselines), patient redress costs, and reputational/trust externalities. At ECR = 0.99, the expected residual fraction is 0.01 — one record in a hundred. The marginal cost of moving from 0.99 to 0.995 under the REAL bound (E·M·R)/(T·S) escalates super-linearly as discovery exhausts declared surfaces and must turn to sampled or forensic enumeration; the crossover point at which marginal containment cost equals marginal expected harm sits near 0.992 in the HIPAA-baseline model. The floor is therefore set at 0.99 with operational targets in the 0.992–0.995 band, reserving the gap as residual operational tolerance.

D.2 PROTECTION (SSPS, floor ≥ 0.97 with adjustment)

PROTECTION governs whistleblower identity, witness records, and fear-of-retribution elimination. Unlike HEALTH, the harm function is non-linear: a single residual instance may be fatal to the protection guarantee. The naive demand is therefore ECR = 1.0, but the REAL bound forbids absolute closure. v1.2 sets the floor at 0.97 with explicit inferential-reconstruction adjustment: if the residual instances are themselves sufficient (singly or combined) to reconstruct identity by inference, the effective floor rises to 0.985 or higher per a published reconstruction model. Below 0.97, no SCALULAR PROTECTION pillar service is offered; between 0.97 and the inferential-reconstruction-adjusted threshold, service is offered with explicit acknowledged residual risk to the protected party.

D.3 LAW and SKILLS_TRADE

LAW (SSLA) and SKILLS_TRADE (SSST) thresholds are set per jurisdiction by COI charter, never below 0.90 for any data class touching personally-identifying material. v1.2 does not derive these centrally because the harm models vary too widely by jurisdiction; the Phase 5 pilot will provide a derivation template.

D.4 Why these floors instead of higher ones

The REAL bound makes ECR = 1.0 unreachable in finite time with finite resources. A floor of 1.0 is therefore not stricter — it is dishonest. The floors in v1.2 are calibrated to the harm-cost crossover under the REAL constraint, so that the threshold lies within the falsifiability surface rather than outside it. A system that promises ECR = 1.0 has implicitly admitted it will not measure.

License and Deposit

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— Joseph A. Sprute, ERES Maestro

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